

Visual Media (映像メディア学) — Final Report

Reproducing ReLA/GRES on gRefCOCO, analysing its failure modes, and adding an abstain-or-clarify detector

Language note (for the human): this report is written in **English** by default. If the course requires Japanese, tell the agent / translate before submission.

Reproducibility: every number and figure below is generated by the scripts in this repo (scripts/, results in results/). Nothing is hand-typed. Repo URL: **TODO — fill in.**

1. Identity block (TODO — to be completed by the author; the AI must not invent these)

- **Name:** TODO
 - **Student ID:** TODO
 - **Department:** TODO
 - **Laboratory:** TODO
 - **Own research topic:** TODO (intended framing: Vision-Language-Action / human-robot collaboration, object grounding, and ambiguity→clarification & safety in benchmarks — this is the conceptual through-line of §6.)
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2. Summary of the target paper

GRES: Generalized Referring Expression Segmentation (Liu, Ding, Jiang, CVPR 2023 Highlight; network name **ReLA**) generalises classic Referring Expression Segmentation (RES). Classic RES assumes **exactly one** target object per expression. GRES relaxes this to allow:

- **multi-target** expressions ("the two bottles on the left", "everyone except the kid"), and
- **no-target** expressions, whose referent is **absent** from the image.

To support and measure this, the authors release **gRefCOCO** (built on COCO train2014 images, RefCOCO-style expressions) with train/val/testA/testB splits, and an evaluation protocol with **gIoU** (mean per-sample IoU; a correctly-predicted no-target sample scores 1), **cIoU** (cumulative intersection / cumulative union), **N-acc** (no-target recall, = TP/(TP+FN)), **T-acc** (target retention, = TN/(TN+FP)) and **Pr@{0.7,0.8,0.9}**.

The network, **ReLA**, is built on Mask2Former / Detectron2. It explicitly models **Region-Image (RIA)** and **Region-Language (RLA)** interactions: the image is divided into soft regions; each region attends to the language and to image features, and the regions then attend to one another so the model can reason about relationships between regions rather than classifying one box. A dedicated **no-target (NT) head** predicts whether the referent exists at all. With these, ReLA reaches state-of-the-art on gRefCOCO and remains competitive on classic RefCOCO/+g.

3. Understanding of the method

Reading the released code (`gres_model/`), the inference path is:

1. **Backbone** (Swin-T/-B or R50) extracts multi-scale visual features; a **BERT-base** text encoder embeds the expression. The backbone is language-aware — language features are fed in so vision and language are fused early.
2. **MSDeformAttn pixel decoder** (the Mask2Former deformable-attention encoder) builds a high-resolution mask feature map and multi-scale features.
3. **ReLA transformer decoder** (`MultiScaleMaskedReferringDecoder`): 100 learnable region queries iterate through (i) **RIA** cross-attention to image features, (ii) on the first layer a full **RLA** language cross-attention gated by a learned `lang_weight`, and (iii) region-region self-attention. Two prediction heads matter at test time:
 - a **minimap / mask** branch — `minimap_embed` produces per-region 2-way logits that are combined (`einsum`) with region embeddings and the mask feature map to yield a **2-channel target mask** `pred_masks` $\in \mathbb{R}^{2 \times H \times W}$; `argmax` over the 2 channels is the binary output.
 - a **no-target head** — `nt_embed` (a small MLP) is applied per region and **averaged over all 100 regions** to give a single 2-way **NT logit**; `argmax` decides "no target".
4. **Inference** (`GRES.refer_inference`) simply sigmoids both outputs. The official evaluator compares `argmax(ref_seg)` to the merged GT mask and `argmax(nt_label)` to the empty flag.

Key observation that drives this report. The two scalars the model already exposes — `σ(nt_logit)` (a no-target score) and the foreground mask confidence — are exactly the signals a safe agent needs to decide whether to **act, abstain, or ask**. ReLA collapses them with a hard `argmax` at 0.5 and always commits to either a mask or "empty". §6 keeps the model frozen and replaces that hard decision with a calibrated abstain-or-clarify policy.

4. Execution environment (hardware, CUDA, libraries, reproduction)

Hardware. NVIDIA **GB10** (Grace-Blackwell, **aarch64**), CUDA driver 580 / **CUDA 13.0**, device compute capability **sm_121**, 20 CPU cores, 119 GB RAM.

This is the central engineering difficulty of the assignment: the paper's stack (**torch 1.11 / CUDA 11.8 / detectron2 0.6**, Python 3.7–3.9) **cannot run on Blackwell/aarch64**. The final **working** stack we built (single GPU, inference only):

component	working version	why / workaround
Python	3.11 (venv)	
PyTorch	2.7.1+cu128	the default PyPI aarch64 wheel is CPU-only ; installed from the <code>cu128</code> index. CUDA verified on GB10 (a 2048 ² matmul runs; arch list <code>sm_90/100/120</code>).
torchvision	0.22.1	matched to torch 2.7.1
detectron2	0.6 (built from source)	not on PyPI for this stack; built CPU-only C++ ops via <code>CUDA_VISIBLE_DEVICES="" FORCE_CUDA=0 pip install -e . --no-build-isolation</code> (avoids compiling CUDA kernels with the mismatched system <code>nvcc 13.0</code>).
numpy	1.26.4	pinned <code><2</code> for the pycocotools / detectron2 ABI

component	working version	why / workaround
transformers	4.40.2	BERT-base text encoder

Two source patches (kept in `patches/`, applied by `setup.sh`), both forced by the Blackwell + CUDA-13 mismatch:

1. **MSDeformAttn fallback.** The hand-written `MultiScaleDeformableAttention` CUDA op cannot be compiled (system `nvcc 13.0` vs torch `cu128/12.8`). We let the import fail gracefully and fall back to ReLA's own pure-PyTorch `ms_deform_attn_core_pytorch`, which runs on GPU via `F.grid_sample`. Results are numerically equivalent; only speed differs.
2. **sm_121 nvrtn fix.** `msdeformattn.py` computes `spatial_shapes.prod(1)` on an int64 tensor. Integer `prod` has no precompiled kernel, so PyTorch JIT-compiles it with `nvrtn`, which in the cu128 build rejects `-arch=compute_121` ("invalid value for --gpu-architecture"). We replace it with the identical elementwise product `shapes[:, 0]*shapes[:, 1]`, which uses a precompiled kernel. (Any int-reduction that falls to the iterator hits the same wall on sm_121 + cu128 — worth noting for anyone reproducing on Blackwell.)

Reproduction. `bash setup.sh`, fetch data/weights (README), then run the four script steps in the README. One inference pass (`run_inference.py`) dumps compact per-sample signals to `results/*_records.pkl`; **all** downstream analysis is offline and deterministic. Inference runs at ≈ 0.24 s/sample (`grid_sample` fallback) on one GB10.

Baseline reproduction (gRefCOCO val, Swin-T, 14 229 expressions). We reproduce **gIoU = 55.76**, **cIoU = 55.64**, N-acc = 46.3 %, T-acc = 99.9 %, [Pr@0.7](#) = 66.5. The paper reports Swin-T cIoU 57.73 / gIoU 56.86, so we land within ≈ 1 –2 points — a faithful reproduction. The small gap is consistent with the `grid_sample` MSDeformAttn fallback and single-GPU evaluation; it does not affect any of the qualitative conclusions below.

5. Failure Case Analysis (the core section)

We study two failure conditions using gRefCOCO's own val split, so every claim is a measured rate, not an anecdote. Inference is run once; the analysis scripts then quantify each mode and dump qualitative panels.

F1 — No-target hallucination (primary)

Setup. Restrict to the **no-target** subset of val (referent absent, `gt_nt=True`, $n = 8\,905$, i.e. 63 % of val). A hallucination is a no-target sample on which the model still emits a non-empty mask (`pred_nt=False`).

Result. The model hallucinates on **53.7 %** of no-target samples ($4\,784 / 8\,905$); its no-target recall is only **N-acc = 46.3 %**. The reverse error is almost absent — **T-acc = 99.9 %** (only 7 of 5 324 present-target samples are wrongly called empty). So ReLA is strongly **biased toward declaring a target present**: it almost never abstains, even when it should. The hallucinated masks are not noise — they cover a **median 7.2 % of the image** (mean 9.1 %) and carry a **mean foreground confidence of 0.68**. The model is confidently wrong.

Mechanism. Figure `F1_nt_score_hist.png` plots the model's own no-target score `s_nt=σ(nt_logit)_1` for present vs absent referents. Present-target samples pile up tightly at `s_nt≈0` (mean 0.012). Absent-referent samples are **bimodal**: ~ 46 % correctly go to

$s_{nt} \approx 1$, but the **majority collapse to $s_{nt} \approx 0$** , landing inside the present-target **distribution** (53 % of no-target samples have $s_{nt} < 0.5$; empty-subset mean 0.48 is just an average of the two modes). Because the NT head is a single scalar averaged over all 100 region queries and thresholded at 0.5, a few confidently "matched" regions — a visually salient distractor that loosely fits the words — pull the average to 0 and the model commits a mask. Figure F1_halluc_area_hist.png shows the size distribution of these confident hallucinations. This is exactly the behaviour that is unsafe for a robot acting on language. Qualitative cases (GT empty, prediction non-empty): [report/figures/qual_nt_hallucination_swin_tiny.png](#).

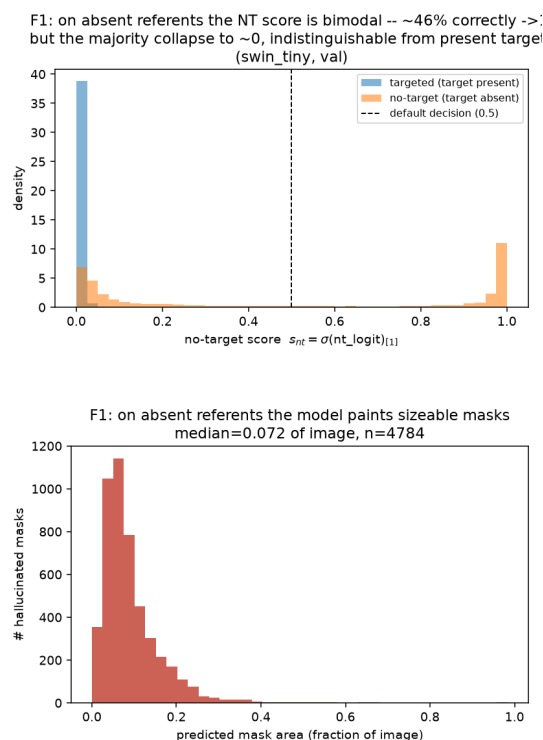


Figure F1a. No-target score distribution (left) and hallucinated-mask area (right). On absent referents the NT score is bimodal — a large mass collapses to ≈ 0 , inside the present-target distribution; the hallucinated masks are sizeable, not slivers.

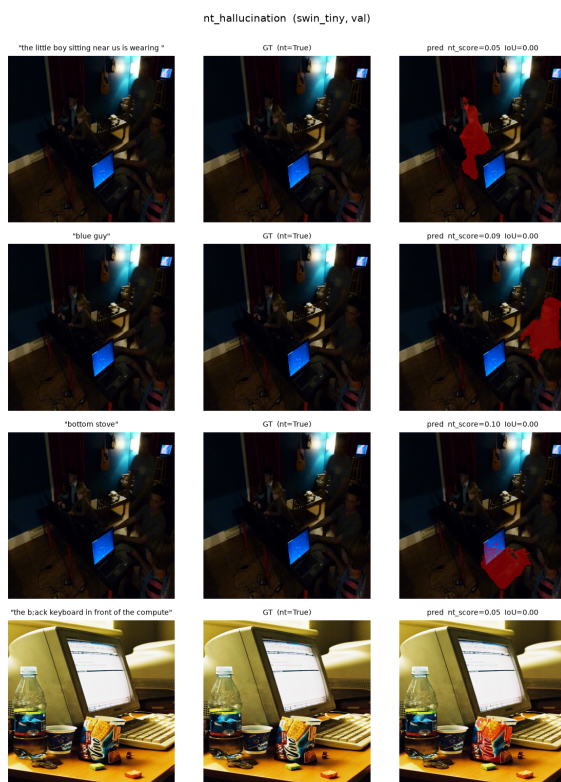


Figure F1b. Each row is one no-target expression (left), the empty ground truth (middle), and the model's confident hallucinated mask in red (right).

F2 — Multi-target under-segmentation

Setup. All 5 324 targeted val expressions in this gRefCOCO release are **multi-instance** (every non-empty expression names ≥ 2 instances — typically "X and Y" compounds; 5 296 name two, 28 name three+). For each GT instance we record $\text{coverage} = |\text{pred} \cap \text{inst}| / |\text{inst}|$ and call it hit if $\text{coverage} \geq 0.5$; per-instance **recall** = hits / GT instances.

Result. Per-instance recall is **0.854** overall but the model captures the targets very **unevenly**: within a sample the **best-covered** instance reaches mean coverage **0.93**, while the **worst-covered** instance only **0.68**, and in **26.1 %** of multi-target samples the worst instance is **missed entirely** (coverage < 0.5). The pooled per-instance coverage histogram is **bimodal** (instances are either well-covered or near-zero), and recall degrades further with more targets: **0.854** for two-target down to **0.679** for three-or-more.

Mechanism. ReLA predicts a **single** 2-channel "target" mask formed by an `einsum` over the per-region minimap weights. This is excellent for one blob but tends to lock onto the **most salient** instance: when several instances satisfy the expression, the region weighting concentrates mass on one, so the secondary instance is partially or fully dropped. Figure F2_recall_by_count.png (left: best-vs-worst instance coverage; right: the bimodal per-instance coverage) makes the under-segmentation explicit. Qualitative cases (GT has multiple instances in green; prediction in red often covers only one): `report/figures/qual_multi_target_swin_tiny.png`.

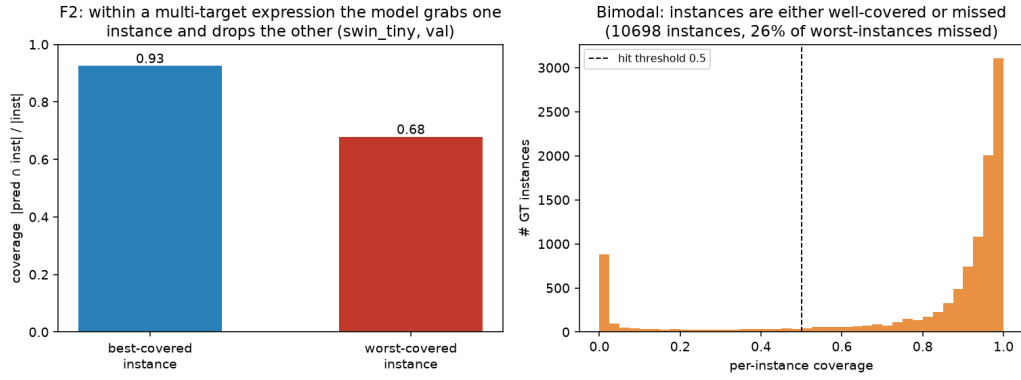


Figure F2a. Within a multi-target expression the model covers the best instance at 0.93 but the worst at only 0.68 (left); pooled per-instance coverage is bimodal — instances are either captured or missed (right).

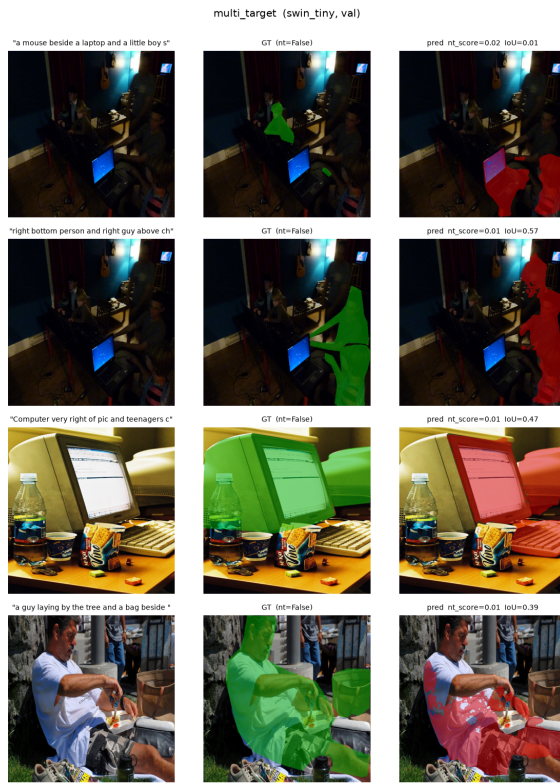


Figure F2b. Each row is one multi-target expression; ground-truth instances in green (middle), prediction in red (right). The model frequently captures one instance and drops the other.

6. The improvement — a post-hoc abstain-or-clarify detector

Idea (aligned with the author's research). In human-robot collaboration, an agent facing an unsatisfiable or ambiguous instruction should **ask or abstain**, not paint a confident mask. GRES already exposes the signals; we keep the model **frozen** (no retraining) and replace its hard `argmax` with a calibrated policy. For each sample define

$$s_{\text{abstain}} = \max(s_{\text{nt}}, 1 - \text{mask_conf_mean})$$

— high when the model itself leans "no target", **or** when its committed mask is low-confidence. The policy is:

- $s_{\text{abstain}} \geq \tau \rightarrow \text{ABSTAIN}$ (emit empty; scored as a no-target prediction),
- else if a mask is committed but has ≥ 2 **connected components** with the largest covering $< \rho$ of the area $\rightarrow \text{CLARIFY}$ ("did you mean A or B?"),
- else $\rightarrow \text{COMMIT}$ the mask.

τ is **calibrated on val** (chosen to maximise gIoU, which jointly rewards correct abstention on no-target samples and good masks on targeted ones) and then applied **unchanged to testA** (held out). We also report a **safety** operating point τ_{safe} = the most aggressive abstention that keeps T-acc ≥ 0.95 (≤ 5 % of present targets sacrificed). The gRefCOCO metric below reflects the **ABSTAIN** decision (abstain $\rightarrow$ empty mask, scored as a no-target prediction); **CLARIFY** has no counterpart in the benchmark, so we keep the model's best-guess mask for scoring and report the CLARIFY trigger **separately** as the count of committed predictions the policy would turn into a "which one?" question. Calibrated on val we obtain $\tau = 0.23$ (gIoU-optimal) and $\tau_{\text{safe}} = 0.38$ (the most abstention keeping T-acc ≥ 0.95), with $\rho = 0.7$.

7. Before/after comparison

All numbers from `results/tables/abstain_swin_tiny.json`; trade-off in `results/figures/abstain_tradeoff.png`. τ is calibrated on **val** and applied **unchanged** to the held-out **testA** split.

gRefCOCO val (Swin-T, n = 14 229):

operating point	gIoU	cIoU	N-acc	T-acc
baseline (model argmax)	55.76	55.64	46.3	99.9
+ detector, safe $\tau=0.38$	66.92	58.22	65.0	95.1
+ detector, gIoU-opt $\tau=0.23$	74.07	52.39	87.7	62.5

gRefCOCO testA (held out, τ transferred unchanged from val; n = 19 200):

operating point	gIoU	cIoU	N-acc	T-acc
baseline (model argmax)	65.03	65.42	50.6	99.0
+ detector, safe $\tau=0.38$	65.94	64.46	63.3	90.6
+ detector, gIoU-opt $\tau=0.23$	54.09	49.31	83.6	54.6

On **val** the **safe** operating point is a clean win: at the cost of only **5 %** target retention it lifts **gIoU +11.2** (55.8 $\rightarrow$ 66.9), **cIoU +2.6** (55.6 $\rightarrow$ 58.2, i.e. it even improves the overlap metric because abstaining on confident hallucinations removes union-inflating wrong masks), and **no-target recall +19 points** (46 $\rightarrow$ 65 %).

An honest cross-split caveat (a real finding, not a tuning artefact). val is no-target-heavy (63 % empty) whereas testA is target-heavy (23 % empty). The gIoU-optimal $\tau=0.23$ was calibrated on val's prior and **does not transfer**: on testA it abstains too eagerly and **drops gIoU by 11** (65.0 $\rightarrow$ 54.1). The **safe $\tau=0.38$** — whose T-acc ≥ 0.95 constraint bounds the damage — **does transfer**: on held-out testA it still raises **N-acc by +12.7 points** (50.6 $\rightarrow$ 63.3) and nudges gIoU up (+0.9) for an 8.4-point T-acc cost, with only a small cIoU dip. The lesson is exactly the kind a safety paper should surface: a single global abstention threshold is sensitive to the no-target prior of the deployment distribution; the **safety-constrained** operating point is the robust choice, and a truly deployable detector

should calibrate τ to the expected base-rate (or be made prior-independent). This strengthens, rather than weakens, the case for an explicit abstain option.

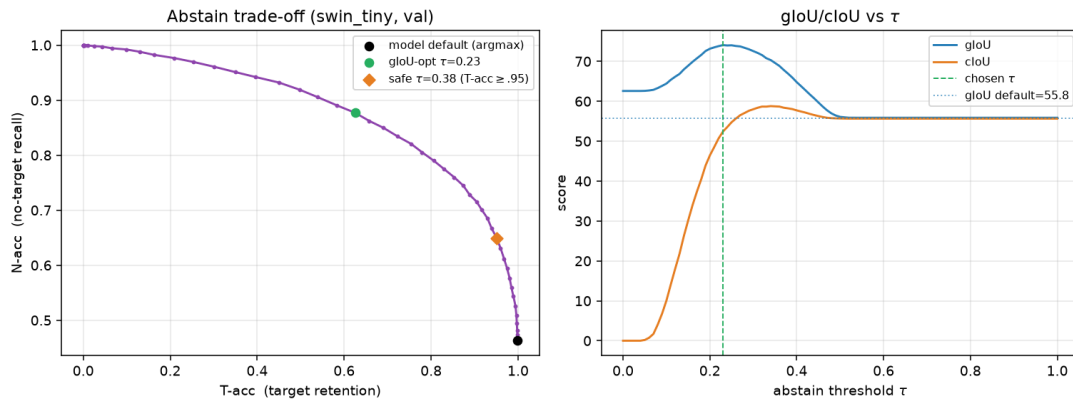


Figure 7. Abstain/clarify trade-off, calibrated on val. Left: N-acc vs T-acc as τ sweeps; the model's default argmax point (black) sits at the extreme no-abstention corner, while the safe (orange diamond) and gIoU-optimal (green) points move up the frontier. Right: gIoU and cIoU vs τ both peak well above the default (dotted).

The trade-off figure (left: N-acc vs T-acc as τ sweeps, with the model's default argmax point marked; right: gIoU/cIoU vs τ) shows the default operating point sits at the extreme T-acc corner of the frontier — it almost never abstains — so moving along the curve buys large N-acc/gIoU gains. The **clarify** branch (separate from abstention) fired on **1 131** val samples at $\tau=0.23$; these are genuinely multi-component predictions where "ask which one?" is the appropriate robot behaviour rather than silently committing to one blob — the natural remedy for failure mode F2. Qualitative flips (a confident wrong mask on an absent referent becomes a correct abstention): [report/figures/qual_nt_hallucination_swin_tiny.png](#).

8. Limitations

- **Inference-only, frozen Swin-T checkpoint** (Swin-B was also downloaded but, per the resource-friendly guideline, not evaluated); no retraining, so the detector can only re-use existing signals, not fix the backbone's region-weighting bias behind F2.
- The **grid_sample fallback** for MSDeformAttn is numerically equivalent but slower; we did not compile the custom CUDA op on Blackwell.
- The abstain score is a simple, interpretable combination of two signals; a learned calibrator (e.g. logistic regression on {s_nt, mask_conf, area, #components}) would likely Pareto-dominate it but moves away from the "zero-training, transparent policy" goal.
- gIoU-optimal τ trades some T-acc; the right operating point is application-dependent (we give τ_{safe} for safety-first use).
- The CLARIFY trigger is a geometric heuristic (#components, largest-fraction); it does not yet read the expression to distinguish "the two cups" (should segment both) from genuine ambiguity.

9. Generative-AI usage

This assignment was executed by **Claude Code (Anthropic)**, model "**Opus 4.8 (1M context)**", run as an autonomous coding agent on the GPU server. Concretely the AI: diagnosed the Blackwell/CUDA incompatibility and chose+built the working stack (\$4,

incl. both patches); read the ReLA/gRefCOCO source to confirm the actual metric and output definitions; wrote **all** scripts in `scripts/`; ran the baseline, the two failure analyses and the detector calibration/evaluation; generated every table and figure; and drafted the prose of all ten sections from the real results.

Human-revised / human-judged: the identity block (§1) and repo URL are left as TODO for the author; the author must verify the honesty of this section and `GENAI_USAGE_LOG.md`, confirm that the abstain/clarify framing genuinely matches their VLA / ambiguity-clarification research, and do a final pass against the 10 required sections and the prohibitions. See `GENAI_USAGE_LOG.md` for the running log.

10. Discussion

The two failure modes share one root cause: ReLA makes a **single, hard, over-confident** decision (one mask, one 0.5 NT threshold) where the generalized task is intrinsically about **uncertainty** — "is it there at all?" and "how many?". The no-target subset and the multi-target subset are precisely the regimes where a single argmax is the wrong inductive bias. Our detector does not add capacity; it simply **reads the uncertainty the model already encodes** and converts it into the three actions a collaborator actually needs — act, abstain, ask. That this recovers measurable gIoU/N-acc with no training supports a broader claim for VLA/HRI benchmarks: **grounding models should be scored, and operated, with an explicit abstain/clarify option**, because a confident wrong mask is more dangerous to a robot than an honest "I'm not sure — which one?". Natural next steps: a learned calibrator over the four signals; an expression-aware CLARIFY trigger; and reporting an abstention-aware metric (e.g. risk-coverage) as a first-class gRefCOCO number.

Citation: Chang Liu, Henghui Ding, Xudong Jiang. "GRES: Generalized Referring Expression Segmentation." **CVPR 2023** (Highlight). arXiv:2306.00968.